

# More than Mortality:

## Early functional outcomes from the ANZICS PROEMS pilot

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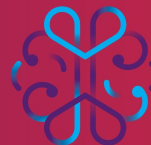
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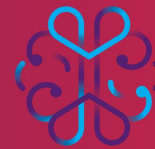
- Refresher on PROEMS
- Overview of the pilot
- Pilot responses
- PROMS
  - PROMS overview
  - PROMs results (so far...)
  - Future directions
- PREMS



# Background: PROEMS

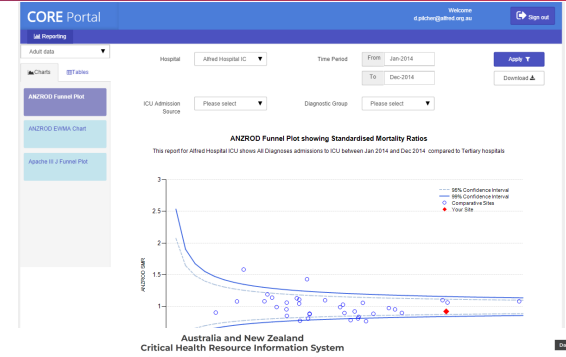


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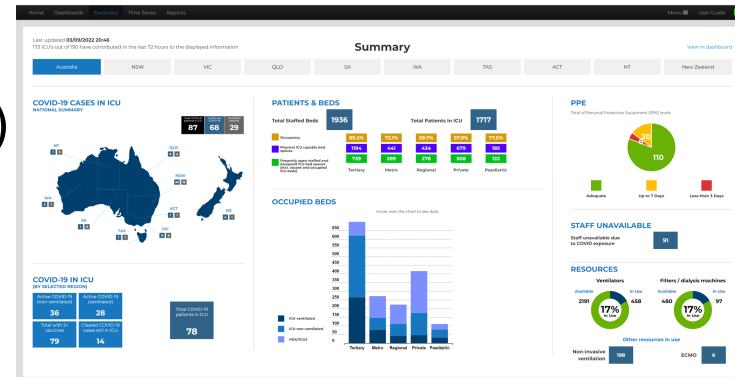
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in the Digital Age  
Merging Hearts and Minds

- We have:
  - Registries for mortality & adverse events
  - Systems to track resource use
- We want to know:
  - Functional outcomes (PROMS)
  - Family experience during ICU (PREMS)
- ...But traditional follow-up is expensive



CHRIS

Australia and New Zealand  
Critical Health Resource Information System



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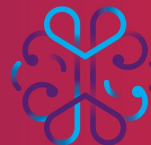
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# Overview: PROEMS pilot

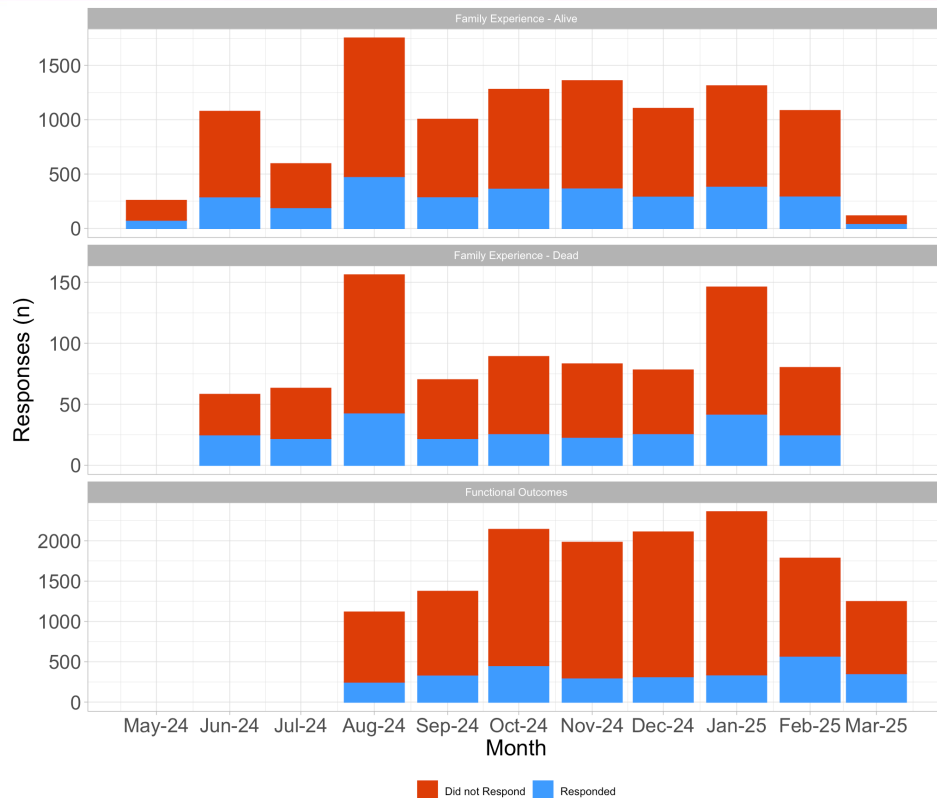


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- Determine feasibility of measuring PROEMS using a registry
- Text-message survey
  - PROM: To patient at 90 days
  - PREM: NoK on ICU discharge
- Aiming ~30,000 delivered surveys
- 16 sites submitting (so far)

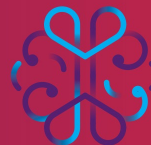


# Overview: Responses

- ~26,000 surveys sent
- ~6,250 responses
- Significant variability in response rate

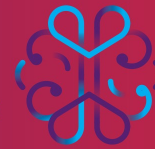


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- **Global Rating of Change**  
Compares before vs. after ICU admission
- **EQ5D-5L**  
Current functional state
  - Self-care (Dressing, washing)
  - Mobility
  - Usual activity (study, work, leisure)
  - Pain
  - Anxiety

Global rating of change

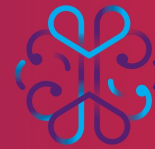
- ☐ Very much worse
- ☐ Somewhat worse
- ☐ A little bit worse
- ☐ Unchanged
- ☐ A little bit better
- ☐ Somewhat better
- ☐ Very much better



# Background: PROMS – EQ5D-5L



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- Raw (domain) scores
- Index Scores:  
Health state on a 0-1 scale
- Value sets:  
Allow population-specific  
rating of health outcomes.

Which of these 2 health states would you prefer to be in for 10 years?  
(all else being equal)

Mobility  
I have no problems in walking about

Anxiety / Depression  
I am extremely anxious or depressed

this one

OR

Mobility  
I have moderate problems in walking about

Anxiety / Depression  
I am not anxious or depressed

this one

they are equal

skip this question for now

**Table 4** Preferred Australian EQ-5D-5L value set

| Dimension          | Level                    | Value set |
|--------------------|--------------------------|-----------|
| Mobility           | 2                        | − 0.039   |
|                    | 3                        | − 0.067   |
|                    | 4                        | − 0.237   |
|                    | 5                        | − 0.242   |
| Self-care          | 2                        | − 0.030   |
|                    | 3                        | − 0.058   |
|                    | 4                        | − 0.213   |
|                    | 5                        | − 0.221   |
| Usual activities   | 2                        | 0         |
|                    | 3                        | − 0.055   |
|                    | 4                        | − 0.162   |
|                    | 5                        | − 0.162   |
| Pain/discomfort    | 2                        | − 0.044   |
|                    | 3                        | − 0.081   |
|                    | 4                        | − 0.276   |
|                    | 5                        | − 0.285   |
| Anxiety/depression | 2                        | − 0.032   |
|                    | 3                        | − 0.066   |
|                    | 4                        | − 0.238   |
|                    | 5                        | − 0.238   |
| N5                 | Any dimension at level 5 | − 0.153   |

## Reference:

Sullivan T, Hansen P, Ombler F, Derrett S, Devlin N. A new tool for creating personal and social EQ-5D-5L value sets, including valuing 'dead.' *Social Science & Medicine*. 2020 Feb 1;246:112707.

Norman R, Mulhern B, Lancsar E, Lorgelly P, Ratcliffe J, Street D, et al. The Use of a Discrete Choice Experiment Including Both Duration and Dead for the Development of an EQ-5D-5L Value Set for Australia. *Pharmacoeconomics*. 2023;41(4):427–38.



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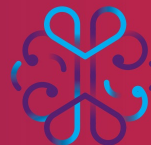


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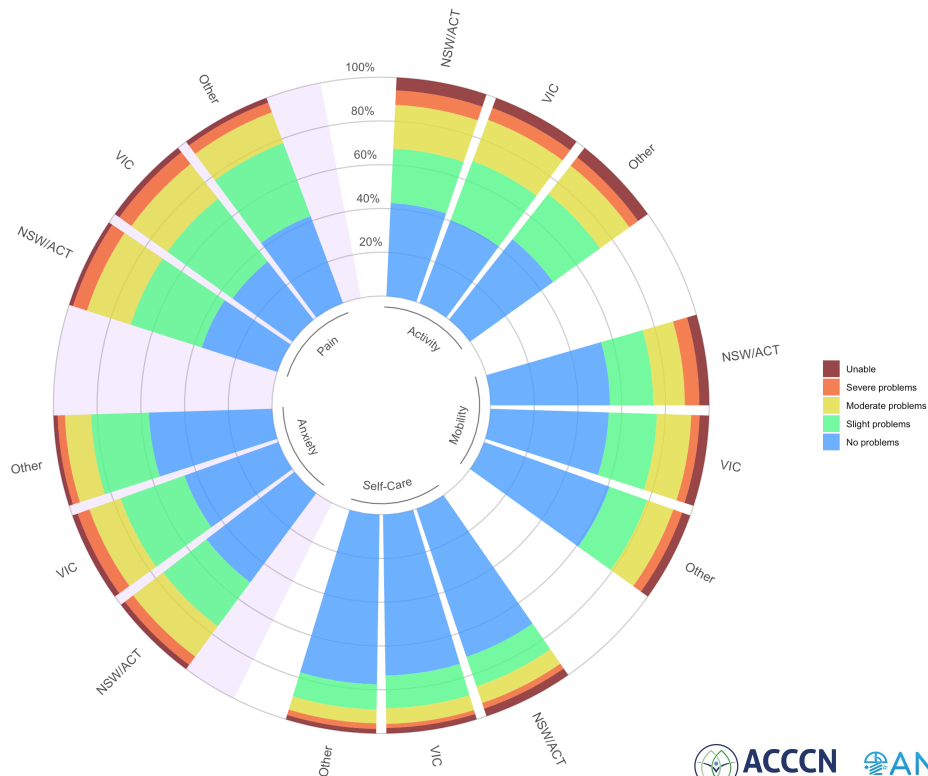
# Functional Outcomes with EQ5D



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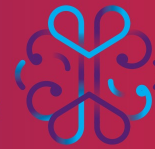


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# Functional Outcomes - Activity

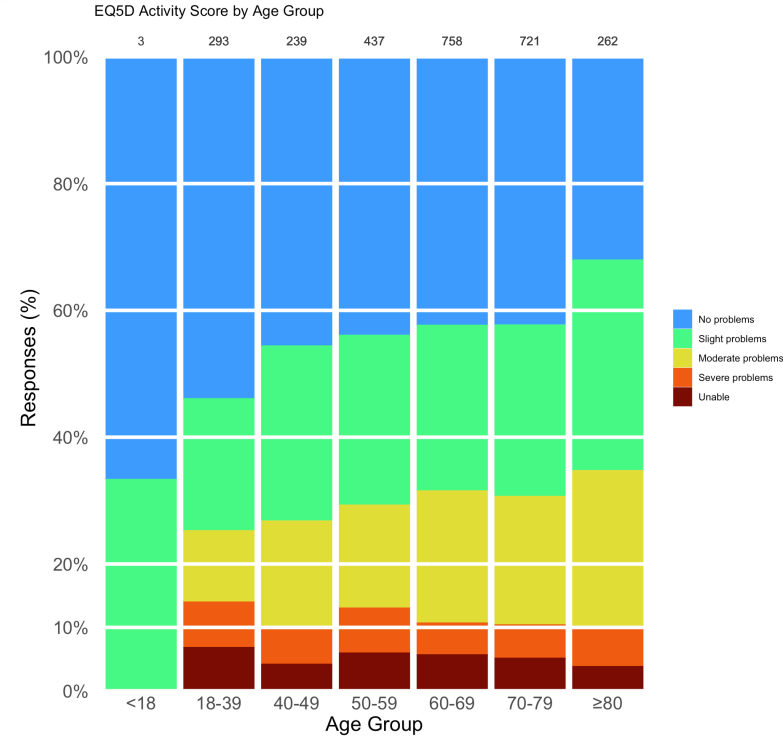


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- Activity appears to be the most sensitive domain to change
- A small but significant number of patients have significant impairment
- Appears consistent across age groups



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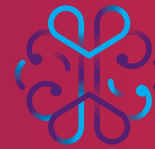


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# Functional Outcomes - Change

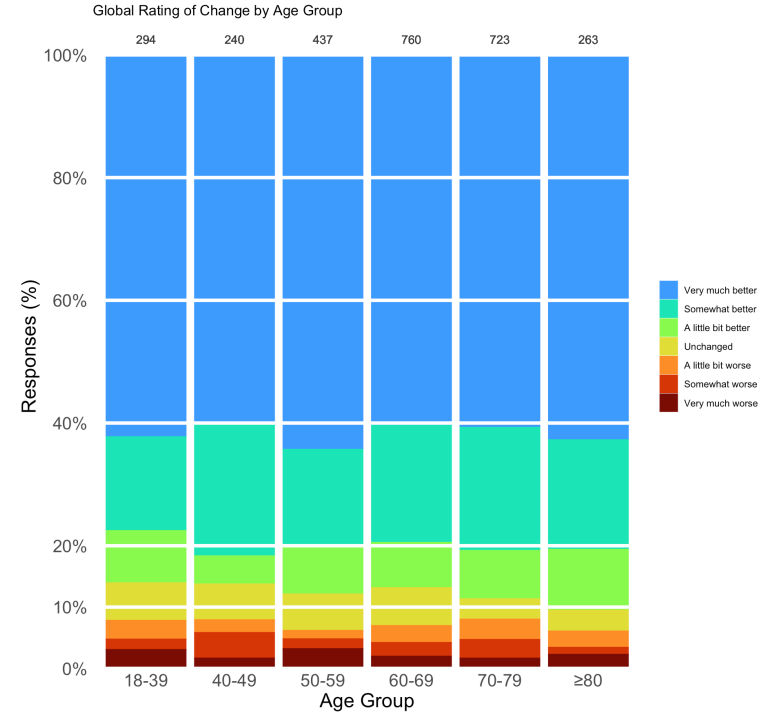


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- Is ICU admission causative?
- Is the disability pre-morbid?
- Both?



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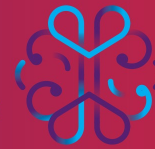


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# Functional Outcomes – Index Scores

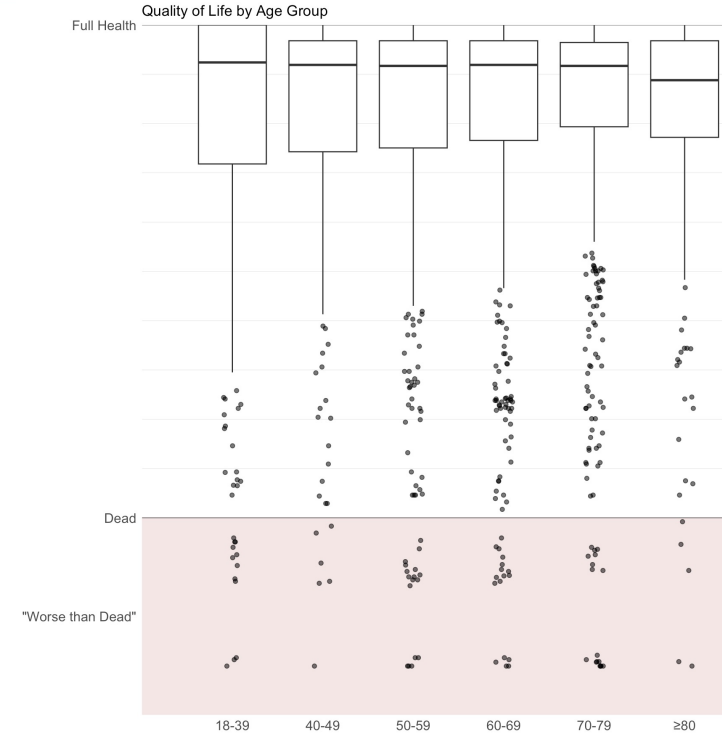


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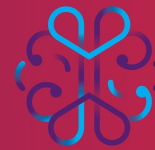
- But most patients do well
- Quality of life remains close to full health for the majority



# Future Directions



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- Everything so far – only PROMS
- Data linkage offers exciting opportunities



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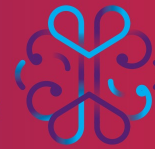


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# Adult Patient Database + PROEMS

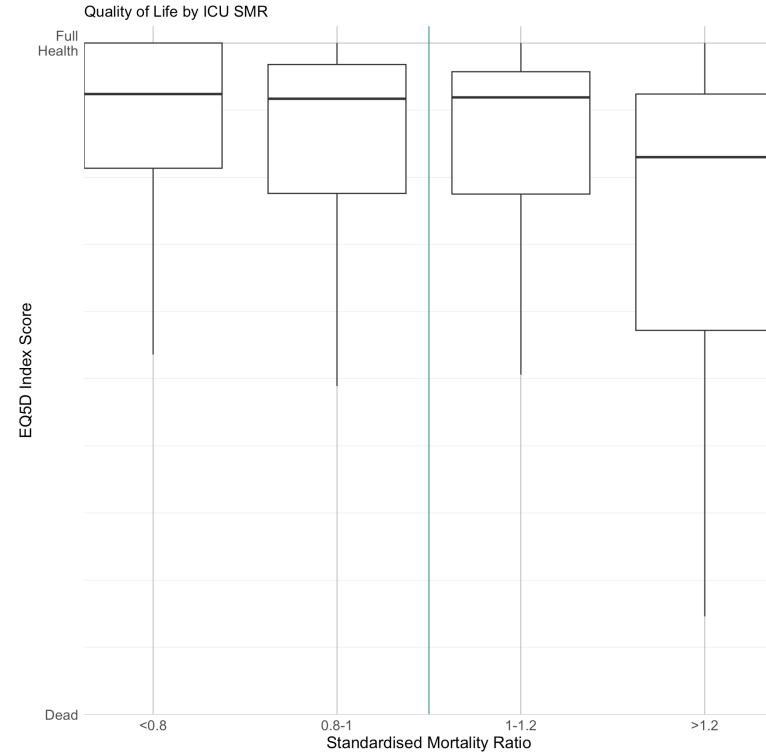


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- Q: Is ICU performance associated with functional outcome?
- A: Yes  
(K-W chi-sq 15.6,  $p = 0.001$ )



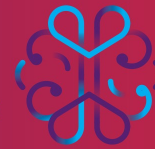
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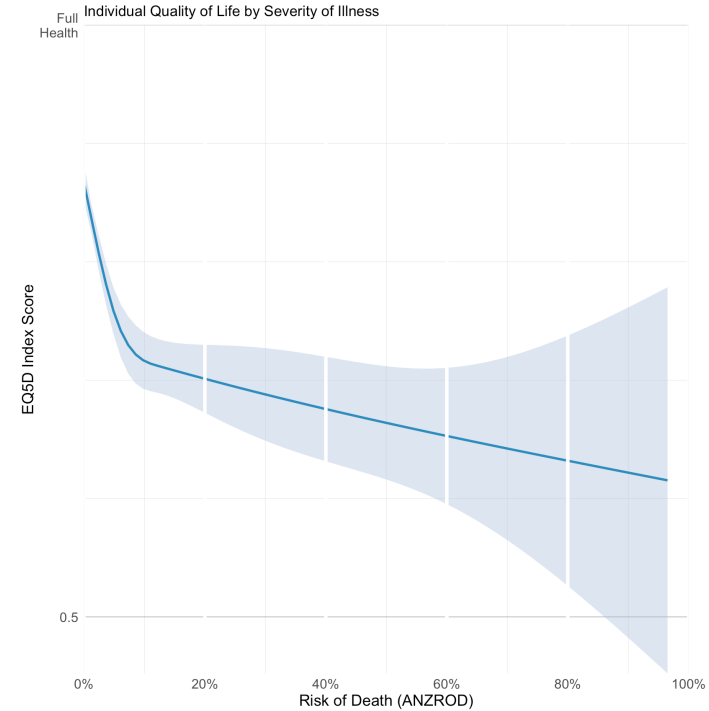
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- Q: Is severity of illness associated with functional outcome?

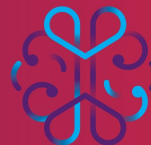
- A: Yes

(GLM;  $p < 2 \times 10^{-16}$ )





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# Thank you!



- Wilson Low
- Shaila Chavan
- Sue Huckson
- Jennifer Hogan
  
- David Pilcher
- Edward Litton
- Paul Secombe

- Canberra Hospital
- Fairfield Hospital
- Liverpool Hospital
- University Hospital Geelong
- Sunshine Hospital
- Gold Coast University Hospital
- Alfred Hospital
- St John Of God Health Care (Subiaco)
- Campbelltown Hospital
  
- North Shore Private Hospital
- Nepean Hospital
- Dunedin Hospital
- Bendigo Health Care Group
- Fiona Stanley Hospital
- Townsville University Hospital
- Wellington Hospital
- Alice Springs Hospital



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